ANALYZING TRENDS BETWEEN UPI USAGE AND CONSUMER EXPENDITURE IN INDIA

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How does the adoption of digital payments via the Unified Payments Interface (UPI) impact Consumer Spending and Final Consumption Expenditure in India, considering factors such as Transaction Volume, Transaction Value, and Bank Affiliation from 2016 to 2023?

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ABSTRACT

This econometric analysis aims to examine the impact of the widespread adoption of UPI digital payments and i ts impact on f inal consumer expenditure in India from 2016 to 2023 . Using both primary (online survey) and secondary data (multivariate regression model), we found that there exists a strong correlation between the two.

Our regression model proved to be statistically valid where our
R Square value equals 0 . 96 , ultimately supporting the f indings found in our primary survey as well.

The f indings of this study reveal the extent to which digital payment methods have influenced consumer expenditure in India, offering insights for other policymakers and f inancial institutions aiming to cultivate a cashless economy.

INTRODUCTION

Launched by the National Payments Corporation of India (NPCI) in 2016 , UPI has emerged as a key driver in the nation' s convergence towards a cashless economy. Its convenient payment capabilities have led to i ts adoption across the country, with the number of transactions skyrocketing over the past few years. This growth reflects the increasing consumer trust in digital payments and the expanding digital infrastructure in India. UPI' s low- cost model, ease of use, and government support have further paved the way for i ts widespread acceptance among both urban and rural populations.

The ability to make instant payments without the need for cash or card has fundamentally changed the way we transact. The shift from traditional payment methods to digital modes is evident in our daily l ives, where UPI transactions have become the default option for many, especially during the COVID- 19 pandemic, which accelerated the digital adoption curve.

To understand i f our postulation applies to our immediate environment, we conducted primary research through a survey amongst users in our demographic, aimed at uncovering how UPI usage has affected consumer spending habits. The survey f indings highlight a significant increase in the frequency of transactions and overall spending behavior among users, suggesting a shift towards a more digital- centric consumption pattern. This primary research forms the foundation of our study, providing f irsthand insights into how digital payments via UPI have influenced consumer behavior.

The rationale behind choosing this topic stems from the need to explore the economic implications of this digital turnover. The broader economic impact of UPI payments remains underexplored.

This research analyzes the influence of two key UPI- related independent variables: transaction volume and transaction value, on Final Consumption Expenditure, while also incorporating an additional variable— Banking Penetration Rate — to account for broader digital adoption trends.

DATA

An overview of the variables we used and why can be found below:

- **Final Consumption expenditure (constant 2015 US$) In India:**
  - Our Dependent Variable (y)
  - Units: in US$ (absolute values)
  - Source: World Bank Open Data
  - Represents the total value of all goods and services consumed by households within an economy. This includes spending on necessities (l ike food, housing, healthcare) as well as discretionary i tems (l ike luxury goods and entertainment). this makes i t a real value adjusted for inflation and a well as rounded and measure of consumer spending.

- **Volume of UPI Transactions in India:**
  - One of the Independent Variables (x_1) Units:
  - In millions
  - Source: National Payments Corporation of India (NPCI). UPI Product Statistics Measures the number
  - of UPI transactions

- **Value of UPI Transactions in India:**
  - One of the Independent Variables (x^2) Units:
  - In millions
  - Source: National Payments Corporation of India (NPCI). UPI Product Statistics Measures the monetary
  - value of UPI transactions

- **Banking Penetration Rate:**
  - One of the Independent Variables (x^3) Units:
  - In millions
  - Source: National Payments Corporation of India (NPCI). UPI Product Statistics Measures the number
  - of active l ive banks affiliated with UPI
  - Selection of this variable was solely accredited to the fact that in order to execute payment, a bank account is required by the individual and i t is directly correlated number of UPI users in the country.

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For our analysis, we utilized both primary and secondary data sources to draw our conclusions. Our primary data was collected through a research survey conducted via Google Forms. This survey aimed to assess whether people are increasingly using UPI for transactions and i f this increased usage has led to a corresponding rise in their spending habits. The insights gained from this survey were crucial in understanding the impact of UPI adoption on consumer behavior.

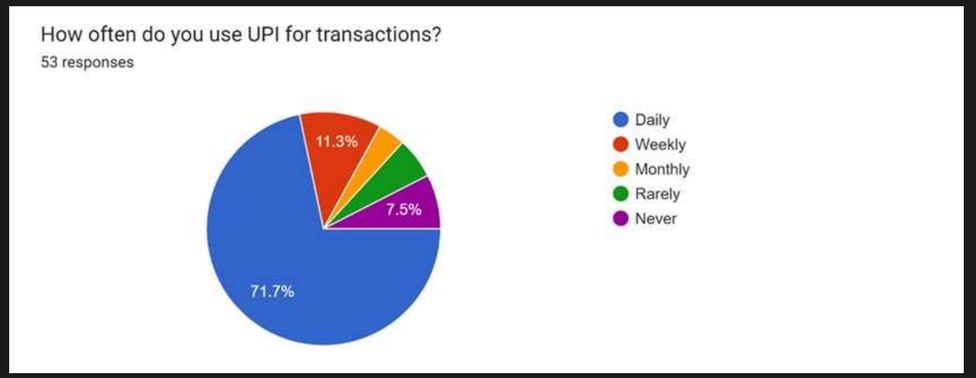
We sourced our secondary data from the **World Bank database** on India as well as **UPI product statistics** data outlined on the NPCI website, both sources can be found in the references section at the end of the presentation.

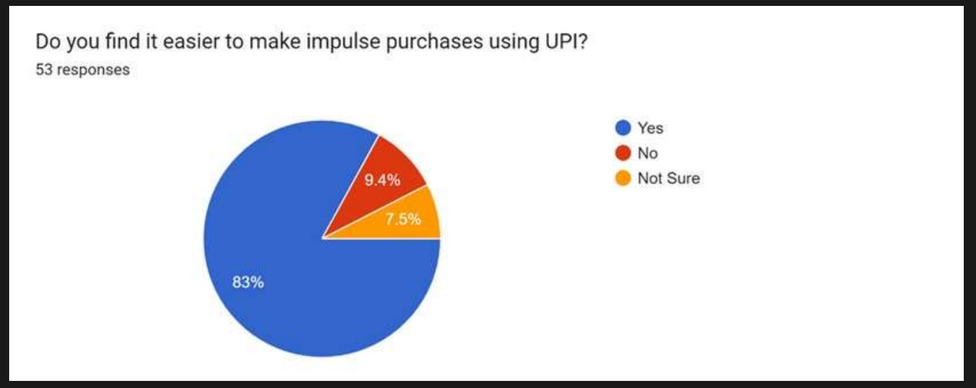
PRIMARY SURVEY

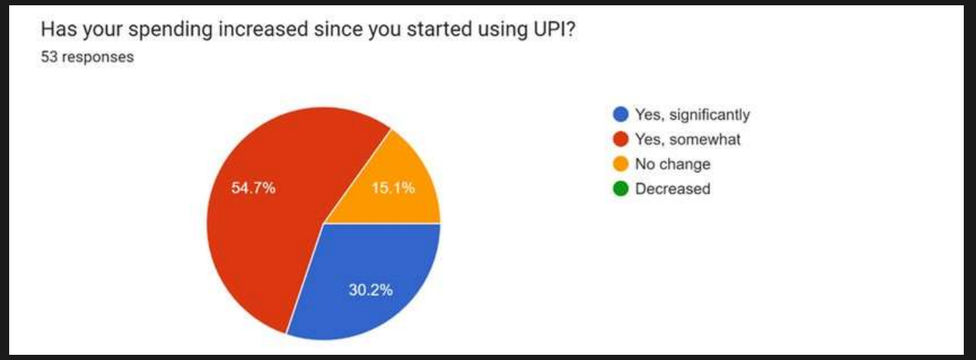
After conducting the primary survey the results indicated that the Convenience of UPI Drives Up Spending in Consumer Sentiment and Behavior according to several respondents, i t has helped to raise expenditure either l i ttle or considerably. Particularly in areas l ike retail buying and food, the accessibility and convenience of usage could encourage more frequent purchases. Although a portion of the participants believe that using UPI has simplified f inancial management, others remain uncertain or hold different opinions. The impact of UPI on f inancial discipline is reflected in a variety of experiences and viewpoints, as evidenced here.

In conclusion , the data indicates that the use of UPI is not just common among younger people but also l inked to higher consumer spending. UPI transactions seem to stimulate larger spending because they are convenient and frequent, especially when i t comes to everyday needs l ike groceries, food, and retail buying. UPI has expedited and streamlined transactions, but some users appear to be using i t less judiciously.

Data collected from the primary survey- Google docs-
https:// docs. google. com/ spreadsheets/ d/ 1 xsq KRz Er8 qx Vl Lh Ef GDWSzq LFW 1 kd H1 v 58 _ 5 p VMw Qaw/ edit?resourcekey=& gid=2034641417 #gid=2034641417







METHODOLOGY

The primary survey was conducted gathering data from 53 respondents Information about the participants' age group, frequency of UPI usage, types of UPI transactions, and opinion on whether UPI has changed their spending patterns were among the many questions they were asked to answer.

After the survey we were able to f ind out that 71 . 7 % of the respondents use UPI on a daily basis, we also found that around 50 . 9 % of the respondents use UPI more than 25 times in a month and around 54 . 7 % of the respondents agree that their spending has increased since they started using UPI

After obtaining information from the World Bank and NPCI, we started developing our multivariate regression model with the goal of examining the positive correlation between UPI usage and consumer expenditure from a wider angle in l ight of all of these f indings and data from the original survey for which the model can be expressed in non stochastic form :

$$\hat{Y}=\beta_0+\beta_1(\text{ UPI volume of transactions})+\beta_2(\text{ UPI value of transactions})+\beta_3(\text{ Number of Banks affiliated with UPI})$$

Where :

$\hat{Y}$ is the dependent variable (Final consumption expenditure (constant 2015 US$))

β_0 is the Intercept

$\beta_1+\beta_2+\beta_3$ are the coefficients for the independent variables.

RESULTS

FĈE=1.54143E+12 -70722682.45 (Transaction Volume) + 144413.6948 (Transaction Value) + 206616527.2 (Banks with UPI)

(49380655305) (53207147808) (77781852.5) (562712.8676) (57541285.27)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.984113313
R Square	0.968479013
Adjusted R Square	0.944838273
Standard Error	49380655305
Observations	8

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	2.99684E+23	9.98948E+22	40.966527	0.001843257
Residual	4	9.7538E+21	2.43845E+21		
Total	7	3.09438E+23			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.54143E+12	53207147808	28.97037413	8.4507E-06	1.3937E+12	1.6892E+12	1.3937E+12	1.6892E+12
vol	-70722682.45	77781852.5	-0.909244	0.41465213	-286679726	145234361	-286679726	145234361
VAL	144413.6948	562712.8676	0.256638337	0.81011747	-1417927.69	1706755.08	-1417927.69	1706755.08
banks	206616527.2	57541285.27	3.59075273	0.02294642	46856307.33	366376747	46856307.33	366376747

1. . R square:
Value= 0 . 9684
Interpretation: 96 % of the variability observed in the dependent variable (Y)
is explained by the independent variable (X).

2. . Significance F:
Value =0 . 001843257
Interpretation: It is much less than 0 . 05 , we have strong evidence to reject the null hypothesis. This states that one of the independent variables we have used in our model has a statistically significant relationship with the dependent variable.

INTERPRETATION OF REGRESSION RESULTS

COEFFICIENTS:

1. **Intercept: 1.54143E+12**

This is the estimated final consumption expenditure (in constant 2015 US$) when the volume of UPI transactions, the value of UPI transactions, and the number of banks affiliated with UPI are all zero. This large positive intercept suggests a significant baseline level of consumption.

2.Volume of Transactions: -70722682.45

For each one-unit increase in the **volume of UPI transactions**, the final consumption expenditure is expected to decrease by about 70.7 million US dollars (in constant 2015 US$), holding the other variables constant. This negative coefficient suggests that as the volume of UPI transactions increases, there might be a substitution effect, where consumers shift their spending patterns, potentially reducing overall consumption.

3. Value of Transactions:144413.6948

For each one-unit increase in the **value of UPI transactions**, the final consumption expenditure is expected to increase by approximately 144,414 US dollars (in constant 2015 US$), holding the other variables constant. This positive coefficient suggests that higher transaction values are associated with increased consumption expenditure, though the effect is relatively small compared to other variables.

4.Banking Penetration:206616527.2

For each additional bank affiliated with UPI, the final consumption expenditure is expected to increase by approximately 206.6 million US dollars (in constant 2015 US$), holding the other variables constant. This suggests that as more banks join the UPI network, it potentially leads to increased consumer spending, possibly due to greater accessibility and convenience.

CONCLUSION

The results of our regression analysis indicate that the model is a good fit for the data and reveals a strong linear relationship between the dependent variable and the independent variables.

Model Fit: The R-squared value of 0.968, which is quite high, suggests that approximately 96.85% of the variability in the final consumption expenditure (dependent variable) is explained by the independent variables in the model. This indicates that our model accounts for a substantial portion of the variation in the dependent variable, demonstrating a high level of explanatory power.

Linear Relationship: The Significance F value of 0.001843257 is significantly below the standard threshold of 0.05, indicating that the model as a whole is statistically significant. This suggests that there is a meaningful linear relationship between the dependent variable and at least one of the independent variables. However, the individual p- values indicate that only the number of banks affiliated with UPI is statistically significant, while the UPI volume and UPI value of transactions are not.

The general trend observed suggests that as UPI transactions increase in volume and value, there is potential for changes in consumer spending habits. Increased transaction volume often reflects higher usage and accessibility of digital payments, which could lead to greater ease of transactions and possibly a shift towards higher consumption.

Banking Penetration Rate: The number of banks affiliated with UPI shows a statistically significant relationship with final consumption expenditure (p value:0.0229) .A higher number of banks offering UPI services likely enhances the accessibility and convenience of digital payments, contributing to more frequent and higher-value transactions.

These findings are consistent with the results from our primary survey, which was designed to be representative of the population. The survey results aligned closely with the dataset used for this regression analysis. The absence of significant discrepancies between the survey results and the population dataset reinforces the validity of our chosen dependent and independent variables.

REFERENCES

- World Bank. (n.d.). *Final consumption expenditure (constant 2015 US$)*. World Bank Open Data. https://data.worldbank.org/indicator/NE.CON.TOTL.KD?end=2023&start=1960&view=chart

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INDIVIDUAL CONTRIBUTIONS:

• DATA ACQUISITION • RESEARCH ON VARIABLES • ABSTRACT • FINAL REGRESSION MODEL • CONCLUSION • PPT	• DATA ACQUISITION • RESEARCH ON VARIABLES • INTRODUCTION • FINAL REGRESSION MODEL • CONCLUSION • PPT	• PRIMARY SURVEY • METHODOLOGY • INTERPRETATION • PPT	• REGRESSION TESTING • METHODOLOGY • INTERPRETATION PPT
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